



$$R'_2 = R_{sc} - R_1$$

$$Z_{sc} = \frac{V_{sc}}{I_{sc}} = \underline{\hspace{2cm}} \Omega$$

$$X_{sc} = X_1 + X'_2 = \sqrt{Z_{sc}^2 - R_{sc}^2} = \underline{\hspace{2cm}} \Omega$$

$$X_1 = X'_2 = \frac{X_{sc}}{2} = \underline{\hspace{2cm}} \Omega$$

No Load Test

$$V_o = \underline{\hspace{2cm}} V, \quad I_o = \underline{\hspace{2cm}} A, \quad W_o = \underline{\hspace{2cm}} W$$

$$\text{Rotational loss, } W_{rot} = W_o - I_o^2 \left(R_1 + \frac{R'_2}{4} \right) = \underline{\hspace{2cm}} W$$

$$\frac{X_m}{2} = \frac{V_o}{I_o} = \underline{\hspace{2cm}} \Omega$$

$$X_m = \underline{\hspace{2cm}} \Omega$$

Predetermination of the steady state performance at rated speed (N=1425rpm)

$$\text{At rated speed, slip } s = \frac{N_s - N}{N_s} =$$

Equivalent circuit at rated speed is given by,

$$\text{Forward impedance, } Z_f = R_f + jX_f = \left(\frac{jX_m}{2} \right) // \left(\frac{R'_2}{2s} + \frac{jX'_2}{2} \right) = \underline{\hspace{2cm}} \Omega$$

$$\text{Backward impedance, } Z_b = R_b + jX_b = \left(\frac{jX_m}{2} \right) // \left(\frac{R'_2}{2(2-s)} + \frac{jX'_2}{2} \right) = \underline{\hspace{2cm}} \Omega$$

$$\text{Total impedance, } Z = (R_1 + jX_1) + Z_f + Z_b = \underline{\hspace{2cm}} \Omega$$

$$\text{Current, } I_1 \angle -\phi = \frac{V_1 \angle 0^\circ}{Z} = \underline{\hspace{2cm}} \Omega$$

$$\text{Power factor} = \cos \phi = \underline{\hspace{2cm}} \text{ lag}$$

$$\text{Input power} = V_1 I_1 \cos \phi = \underline{\hspace{2cm}} W$$

$$\text{Forward torque, } T_f = I_1^2 R_f = \underline{\hspace{2cm}} \text{ sync. watts}$$

$$\text{Reverse torque, } T_b = I_1^2 R_b = \underline{\hspace{2cm}} \text{ sync. watts}$$

$$\text{Resultant torque, } T = T_f - T_b = \underline{\hspace{2cm}} \text{ sync. watts}$$

$$\text{Mechanical power developed } P_{mech} = (1-s)T = \underline{\hspace{2cm}} W$$

$$\text{Output power} = P_{mech} - W_{rot} = \underline{\hspace{2cm}} W$$

$$\text{Resultant torque in N-m} = \frac{P}{2\pi N_s/60} = \underline{\hspace{2cm}} N-m$$

$$\text{Efficiency} = \frac{\text{output}}{\text{input}} \times 100 = \underline{\hspace{2cm}} \%$$

RESULT

- No load and blocked rotor tests on single phase induction motor were conducted
- Equivalent circuit parameters determined and its performance at rated speed determined from equivalent circuit